

Name : YASHVI MEHTA

Roll no. : 231071079

LY BTech Computer

Batch D

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Cross Platform App Dev Lab 4

AIM -

To install the Flutter/Dart development environment and develop a mobile application that displays an interactive map, shows the user's current location, and places markers on the map using the Google Maps platform.

THEORY -

Flutter is an open-source UI framework used to develop cross-platform mobile applications using the **Dart programming language**. Dart provides the programming language and Flutter provides the framework, widgets, and tools required to build the application's user interface.

Dart/Flutter does not have built-in map functionality. Interactive maps can be integrated using external packages available through **pub.dev**. The Google Maps Flutter plugin provides a Flutter widget that connects with the native Google Maps SDK and allows an interactive Google Map to be displayed inside a Flutter application.

1. Flutter Map Widget

The map is integrated into the Flutter application's user interface as a widget. Like other Flutter widgets, it can be placed inside the layout and configured with properties such as the initial camera position and markers. This allows the map to become an interactive part of the application rather than a separate screen.

2. Camera Position

The camera determines which portion of the map is initially visible to the user. It consists of a geographical location along with a zoom level and can also include tilt

and bearing. When the user's location is obtained, the camera can be moved and zoomed to focus on that location.

3. Markers

Markers are graphical pins placed at specific latitude and longitude coordinates. They are useful for identifying the user's current position and other points of interest. A marker can contain a title and description that are displayed when the user interacts with it. Multiple markers can be supplied to the map and displayed simultaneously.

4. Location Permission and Geolocation

Since the application accesses the device's location, the user must grant location permission. After permission is granted, the application requests the current latitude and longitude from the device's location service. These coordinates can then be used to update the map and show the user's position.

5. API Key and Google Cloud

Google Maps services require an API key associated with a Google Cloud project. The required Maps SDK is enabled in the project, and the generated key is configured in the Flutter application's native Android or iOS configuration. API-key restrictions can also be applied to limit unauthorized usage.

6. External Packages

Flutter applications can use packages from **pub.dev** to add functionality that is not provided directly by Dart or Flutter. For a map application, a Google Maps package is used to render the map, while a location package can be used to obtain the device's geographical coordinates. These packages are declared as project dependencies and downloaded before running the application.

7. User Interaction

An interactive map allows users to drag the map to view different areas and use pinch gestures to zoom. Marker interaction can display additional information, while a floating action button can re-center the map on the user's current position. These interactions make the application more useful for location-based tasks.

8. Application Flow

The general working of the application can be described as:

Launch Application → Load Map → Request Location Permission → Obtain Current Location → Update Camera → Display User Location and Markers → Allow User Interaction.

This workflow combines the map, location, permission, and marker components into a single location-aware Flutter application.

Thus, a Flutter map application combines **Flutter widgets, Google Maps services, geolocation, API authentication, and device permissions** to create a location-aware interactive application. The same concepts can be extended to more advanced features such as route drawing, place search, and geofencing.

APP OVERVIEW -

This is an interactive map app for your phone that helps you see where you are and discover nearby landmarks.

Shows an Interactive Map: When you open the app, you see a full-screen map that you can pan, pinch, and zoom just like Google Maps.

Finds Your Location: The app asks for permission to access your phone's GPS. Once allowed, it automatically finds your exact position, glides the map over to you, and drops a blue pin marked "You are here."

Highlights Nearby Landmarks: On the map, you will see pins for popular nearby spots (such as VITI college, Five Gardens, and Dadar Station). Tapping any pin pops up a small label with its name and a brief description.

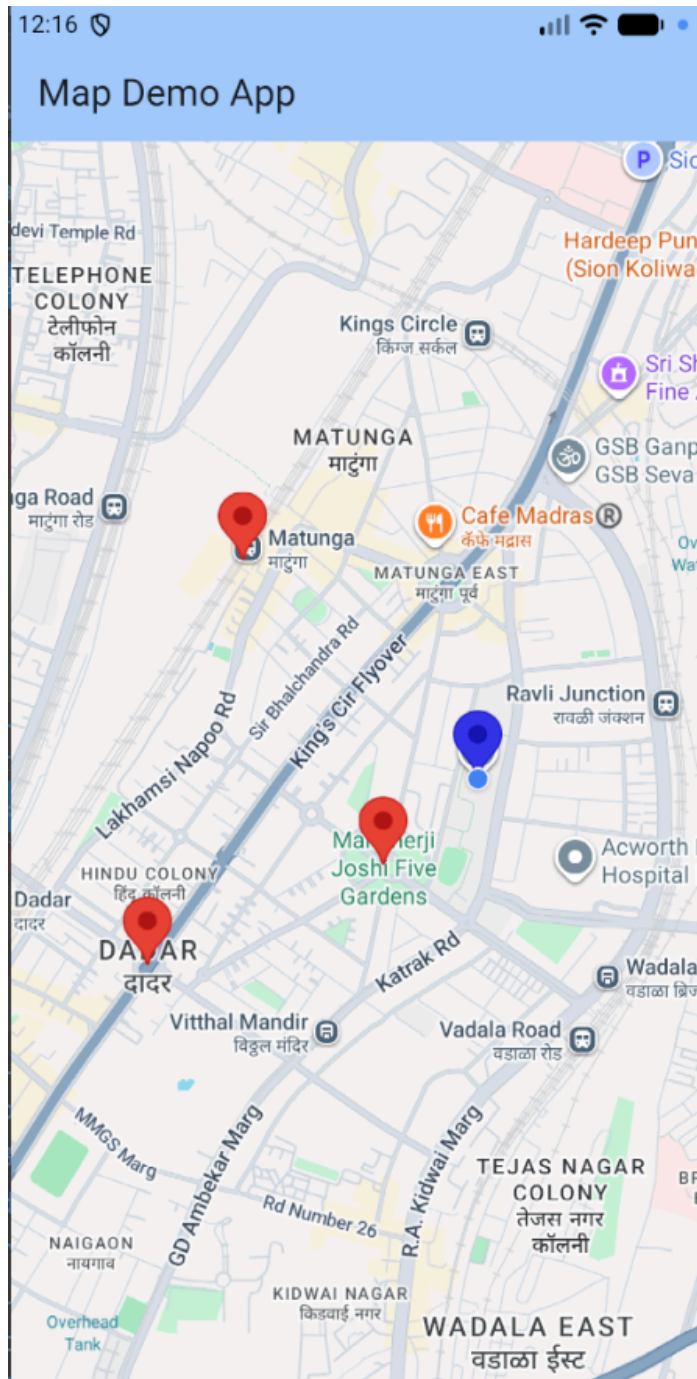
One-Tap "Find Me" Button: If you scroll around and explore other areas, you can tap the floating target button in the bottom corner at any time to instantly zoom back to where you are standing.

Friendly Warnings: If your phone's GPS is turned off or if you decline the location permission, the app won't crash it gently informs you with a short message on the screen so you know why your location isn't showing.

REPO LINK FOR CODE -

https://github.com/YashviMehta03/CPAD_lab_repos/tree/main/maps_app

SCREENSHOTS -



Device's current location -



3 Markers -



CONCLUSION -

The experiment successfully demonstrated the development of a location-aware map application using Dart and Flutter. The application integrated Google Maps to display an interactive map, obtained the device's current location using location services, and displayed custom markers at specified coordinates. The experiment also provided an understanding of API key configuration, package integration, permissions, and user interaction with the map. Overall, the experiment established the basic concepts required for developing Flutter applications with map and location-based features.